Comparative Study of No-Code and Low-Code Development Using Zoho Creator

1. Aim

To study and compare **No-Code and Low-Code application development approaches** using **Zoho Creator**, and to understand how both approaches can be used to develop business applications efficiently with minimum traditional programming.

2. Objectives

1. To understand the concepts of No-Code and Low-Code development.
2. To study the features of the Zoho Creator development platform.
3. To develop a simple application using the No-Code approach.
4. To understand how Low-Code features such as Deluge scripting can extend application functionality.
5. To compare No-Code and Low-Code development based on ease of use, coding requirements, flexibility, development time, and customization.
6. To identify suitable applications for each development approach.

3. Introduction

Traditional software development requires programmers to write a significant amount of source code. **No-Code** and **Low-Code** development platforms reduce this requirement by providing visual development tools, pre-built components, forms, workflows, databases, and automation features.

Zoho Creator is a low-code application development platform that allows users to build business applications using visual development tools. It supports both **No-Code development**, where applications can be created primarily through configuration and drag-and-drop components, and **Low-Code development**, where developers can use scripting such as **Deluge** to implement customized business logic.

4. Theory

4.1 No-Code Development

No-Code development allows users to create applications with little or no programming.

Typical features include:

- Drag-and-drop interface design
- Pre-built forms and fields
- Visual workflows
- Built-in reports and dashboards
- Configuration-based application logic
- Database creation through forms
- Ready-to-use application components

Advantages:

- Easy for beginners
- Requires minimal programming knowledge
- Faster application development
- Reduces development effort
- Suitable for simple business applications

Limitations:

- Limited customization
- Complex business logic can be difficult
- Dependency on platform features
- Less control over application behaviour

4.2 Low-Code Development

Low-Code development combines visual development with a limited amount of programming.

In Zoho Creator, developers can use **Deluge (Data Enriched Language for the Universal Grid Environment)** to add customized logic to applications.

Examples include:

- Custom calculations
- Conditional operations
- Automated notifications
- Data validation
- Record manipulation

- Integration with external services
- Customized workflows

Advantages:

- More flexible than No-Code
- Faster than traditional programming
- Supports complex business logic
- Allows integration and customization
- Suitable for medium and complex applications

Limitations:

- Requires some programming knowledge
- More difficult than pure No-Code development
- Developers need to understand the platform's scripting environment

5. Software/Platform Required

- Zoho Creator
- Web browser
- Internet connection
- Zoho account
- Basic understanding of databases and application development
- Deluge scripting knowledge for Low-Code implementation

6. Application Selected

For comparison, a simple **Online Food Ordering Application** can be developed using Zoho Creator.

The application can contain:

- Customer registration
- Customer details
- Food item details
- Restaurant details
- Order form
- Order status
- Payment details
- Reports and dashboards

7. Procedure

Part A – No-Code Development

Step 1:
Open Zoho Creator and create a new application.

Step 2:
Create forms using the visual form builder.

Example forms:

- Customer Form
- Restaurant Form
- Food Item Form
- Order Form

Step 3:
Add fields such as:

- Customer Name
- Phone Number
- Food Item
- Quantity
- Price
- Order Date
- Order Status

Step 4:
Create relationships between the required forms.

Step 5:
Create reports to display customer orders and food items.

Step 6:
Create a dashboard to display important information such as total orders and order status.

Step 7:
Use the visual workflow tools to automate basic operations without writing scripts.

Part B – Low-Code Development

Step 1:
Use the same application created using the No-Code approach.

Step 2:
Open the workflow or required application component where customized logic is needed.

Step 3:
Add Deluge scripting to implement business rules.

For example, when a customer selects a food item and enters the quantity, a script can calculate the total price.

Step 4:
Implement conditional logic.

Example:

- If quantity > 5, apply a discount.
- If the order is confirmed, update the order status.
- If the order is cancelled, update the corresponding record.

Step 5:
Create automated notifications using workflow and scripting features.

Step 6:
Test the application with different input values.

Step 7:
Generate reports and dashboards to verify the results.

8. Comparison of No-Code and Low-Code

Parameter	No-Code	Low-Code
Programming knowledge	Very little or none	Basic programming knowledge
Development method	Visual configuration	Visual tools + scripting
Development speed	Very high	High
Customization	Limited	High
Complexity supported	Simple to moderate	Moderate to complex
Flexibility	Lower	Higher
Maintenance	Easy	Moderate
Business-user friendly	Excellent	Good
Coding requirement	Minimal/none	Limited
Zoho Creator example	Forms, reports and basic workflows	Deluge scripts, custom logic and integrations
Suitable for	Simple business applications	Customized and complex applications

9. Observation

The No-Code approach allows an application to be created quickly using forms, reports, dashboards, and visual workflows. It is suitable when the application's requirements can be achieved using the built-in features of Zoho Creator.

The Low-Code approach provides greater flexibility because customized logic can be implemented using Deluge scripting. It is more suitable when the application requires calculations, conditional operations, automation, integrations, or advanced business rules.

10. Result

A comparative study of **No-Code and Low-Code development using Zoho Creator** was successfully carried out.

The study showed that:

- **No-Code** is easier and faster for developing simple applications.
- **Low-Code** provides greater customization and flexibility.
- Zoho Creator supports both visual development and scripting-based development.
- No-Code is suitable for users with little programming knowledge.
- Low-Code is suitable for developers who require additional control over application behaviour.

11. Conclusion

No-Code and Low-Code development approaches significantly reduce the time and effort required to develop applications. **Zoho Creator** provides a suitable environment for studying both approaches because applications can first be created using visual tools and then enhanced with Deluge scripting.

Therefore, **No-Code is preferable for simple and rapidly developed applications**, while **Low-Code is preferable when additional customization, automation, and complex business logic are required**. In practical development, both approaches can be combined to achieve faster development while maintaining sufficient flexibility.